

# Benjamin (Ben) Rachunok

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## Academic Appointments

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### Assistant Professor

*North Carolina State University, Raleigh, NC*

### Industrial and Systems Engineering

2023–

### Graduate Faculty

*North Carolina State University, Raleigh, NC*

### Operations Research

2023–

### Rising Environmental Leadership Program Fellow

*Stanford University, Stanford, CA*

### Woods Institute for the Environment

2020–2021

### R&D Intern

*Sandia National Labs, Albuquerque, NM*

Supervisor: Dr. Andrea Staid

### Center for Computing Research

2017–2020

## Education & Training

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### Postdoctoral Researcher

*Stanford University, Stanford, CA*

Advisor: Dr. Sarah Fletcher

### Civil and Environmental Engineering

2020–2022

### Ph.D.

*Purdue University, West Lafayette, IN*

Advisor: Dr. Roshanak Nateghi

### Industrial Engineering

2016–2020

### Bachelor of Science

*North Carolina State University, Raleigh, NC*

### Industrial and Systems Engineering

2011–2016

## Research Support

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### Collaborations in Artificial Intelligence and Geosciences (CAIG)

\$389,206 NC State / \$884,606 total, Co-PI

Artificial Intelligence-Driven Optimization of Coastal Morphodynamics and Infrastructure Resilience

Collaborative PIs: Katherine Anarde (NC State), Brad Murray (Duke), Laura Moore (UNC-Chapel Hill)

### National Science Foundation

August 2026–July 2029

### Infrastructure Systems and People

\$80,054 NC State / \$697,903 total, PI

AI-Enabled Decision Support for Rate Setting and Infrastructure Investment in Decentralized Water Utilities under Drought

Collaborative PIs: Marta Zaniolo (Duke), Eric Edwards (UC Davis), Mehdi Nemati (UC Riverside)

### National Science Foundation

September 2026–August 2029

**North Carolina Collaboratory**  
\$99,820, *Lead PI*  
North Carolina Data Center Water Use Analysis

**State of North Carolina**  
May 2025-December 2026

**North Carolina Collaboratory**  
\$99,719, *Lead PI*  
North Carolina Power System Restoration Optimization

**State of North Carolina**  
May 2025-December 2026

**Digitizing Utilities Prize, Phase 1**  
\$75,000, *Lead PI*  
North Carolina Short-term Power Outage Prediction

**Department of Energy**  
June 2024-June 2025

**R.L. Rabb Science and Society Fund**  
\$7625, *Lead PI*  
North Carolina Climate & Health Systems Impacts Symposium

**North Carolina State University**  
June 2024-June 2025

**Research Collaboration Fund**  
\$30,000, *Lead NC State PI*  
Leveraging AI for Real-time Prediction of Extreme Weather Events  
to Foster Climate Resilience: a Global Research Alliance

**University Global Partnership Network**  
August 2024-August 2025

**UPS Endowment Fund**  
\$100,000, *Co-PI*  
Assessing Drought Impacts on Urban Water Affordability and Access

**Stanford University**  
September 2022-August 2023

## Peer-Reviewed Journal Articles

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\* indicates equal contributions, Underline indicates an advisee

- [29] Y. Essus, **B. Rachunok**, T. Logan, K. Hardaway, and D. Bristow. "Towards Operationalizing Ecological Resilience for Infrastructure Decision-Making". In: *Risk Analysis* (2026). doi: 10.1111/risa.70316.
- [28] A. Helmrich, J. R. Wolfenbarger, K. Hardaway, **B. Rachunok**, and M. Haraguchi. "Flexible infrastructure design for changing conditions". In: *Environmental Research: Infrastructure and Sustainability* (2026). doi: 10.1088/2634-4505/ae9c6f.
- [27] T. Logan, **B. Rachunok**, K. Hardaway, D. Bristow, A. Reilly, D. Johnson, J. Johansson, G. Cremen, J. Boakye, and P. Schweizer. "Advancing systemic risk assessment for complex, interdependent systems: a research agenda". In: *Risk Analysis* 46.5 (2026). doi: 10.1111/risa.70243.
- [26] Y. Essus and **B. Rachunok**. "A Computational Framework for Modeling Access to Services and the Resilience Impacts of Electric Vehicles During Blackouts". In: *IEEE Access* (2025). doi: 10.1109/ACCESS.2025.3629567.
- [25] K. Hardaway, K. Best, and **B. Rachunok**. "Leveraging the American Housing Survey to Quantify Risk and Resilience". In: *Risk Analysis* (2025). doi: 10.1111/risa.70100).
- [24] Y. Essus and **B. Rachunok**. "Electric Vehicles Limit Equitable Access to Essential Services During Blackouts". In: *NPJ Urban Mobility* (2024). doi: 10.1038/s44333-024-00007-3.
- [23] **B. Rachunok**, A. Verma, and S. Fletcher. "Predicting and understanding residential water use with interpretable machine learning". In: *Environmental Research Letters* 19.1 (2024). doi: 10.1088/1748-9326/ad1434.
- [22] J. Skerer, A. Verma, M. Edwards, **B. Rachunok**, and S. Fletcher. "Alternative household water affordability metrics using water bill delinquency behavior". In: *Environmental Research Letters* (2024). doi: 10.1088/1748-9326/ad5609.

- [21] A. Nayak, **B. Rachunok**, B. Thompson, and S. Fletcher. "Socio-hydrological impacts of rate design on water affordability during drought". In: *Environmental Research Letters* 18.12 (2023). doi: 10.1088/1748-9326/ad0994.
- [20] **B. Rachunok** and S. Fletcher. "Socio-hydrological drought impacts on urban water affordability". In: *Nature Water* 1.1 (2023). doi: 10.1038/s44221-022-00009-w.
- [19] J. Bennett, **B. Rachunok**, R. Flage, and R. Nateghi. "Mapping climate discourse to climate opinion: An approach for augmenting surveys with social media to enhance understandings of climate opinion in the United States". In: *PLOS ONE* (2021). doi: 10.1371/journal.pone.0245319.
- [18] M. Choi, **B. Rachunok**, and R. Nateghi. "Short-term solar irradiance forecasting using convolutional neural networks and cloud imagery". In: *Environmental Research Letters* 16.4 (2021). doi: 10.1088/1748-9326/abe06d.
- [17] R. Obringer, **B. Rachunok**, D. Maia-Silva, M. Arbabzadeh, R. Nateghi, and K. Madani. "The overlooked environmental footprint of increasing Internet use". In: *Resources, Conservation and Recycling* 167 (2021). doi: 10.1016/j.resconrec.2020.105389.
- [16] **B. Rachunok**, J. Bennett, R. Flage, and R. Nateghi. "A path forward for leveraging social media to improve the study of community resilience". In: *International Journal of Disaster Risk Reduction* 59 (2021). doi: 10.1016/j.ijdr.2021.102236.
- [15] **B. Rachunok** and R. Nateghi. "Overemphasis on recovery inhibits community transformation and creates resilience traps". In: *Nature Communications* 12.1 (2021). doi: 10.1038/s41467-021-27359-5.
- [14] N. Alemazkoor, **B. Rachunok**, D. R. Chavas, A. Staid, A. Louhghalam, R. Nateghi, and M. Tootkaboni. "Hurricane-induced power outage risk under climate change is primarily driven by the uncertainty in projections of future hurricane frequency". In: *Scientific reports* 10.1 (2020). doi: 10.1038/s41598-020-72207-z.
- [13] R. Kumar \*, **B. Rachunok** \*, D. Maia-Silva, and R. Nateghi. "Asymmetrical response of California electricity demand to summer-time temperature variation". In: *Scientific Reports* (2020). doi: 10.1038/s41598-020-67695-y.
- [12] A. Paulvannan Kanmani, R. Obringer, **B. Rachunok**, and R. Nateghi. "Assessing Global Environmental Sustainability Via an Unsupervised Clustering Framework". In: *Sustainability* (2020). doi: 10.3390/su12020563.
- [11] **B. Rachunok** and R. Nateghi. "The sensitivity of electric power infrastructure resilience to the spatial distribution of disaster impacts". In: *Reliability Engineering & System Safety* (2020). doi: 10.1016/j.res.2019.106658.
- [10] **B. Rachunok**, A. Staid, J.-P. Watson, and D. L. Woodruff. "Assessment of wind power scenario creation methods for stochastic power systems operations". In: *Applied Energy* 268 (2020). doi: 10.1016/j.apenergy.2020.114986.
- [9] **B. Rachunok**, J. Bennett, and R. Nateghi. "Twitter and Disasters: A Social Resilience Fingerprint". In: *IEEE Access* (2019). doi: 10.1109/ACCESS.2019.2914797.

## Manuscripts Under Review

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\* indicates equal contributions, Underline indicates an advisee

- [8] J. Chala Lancheros and **B. Rachunok**. "Prepare vs Response: Stochastic Modeling of Flood Resilience in NC Rural Schools". In: *Risk Analysis* (2026).
- [7] T. Groton and **B. Rachunok**. "Online Grocery Reviews Enable Population-Level Measurement of Perceived Food Access". In: *Nature Food* (2026).

- [6] E. Harris and **B. Rachunok**. "Clustering-based Assessment of Power System and Energy System Interdependence". In: *Reliability Engineering and System Safety* (2026).
- [5] P. Leavitt, F. Livingston, B. McConnell, and **B. Rachunok**. "Anticipating Household Rescue Demand in Hurricanes Using Socio-Demographic Data and Machine Learning". In: *International Journal of Disaster Risk Reduction* (2026).
- [4] R. Smith, B. McConnell, R. King, and **B. Rachunok**. "Two for One: Multi-Objective Disaster Recovery Models Improve Public Utility Post Disaster". In: *International Journal of Disaster Risk Reduction* (2026).
- [3] R. Smith, **B. Rachunok**, B. McConnell, and R. King. "Fast is Fair: Trade-offs in Cell Tower Network Restoration". In: *Reliability Engineering and System Safety* (2026).
- [2] A. Biswas, A. Helmrigh, and **B. Rachunok**. "Flood-related Tweets offer High-resolution impact assessments". In: *npj Urban Sustainability* (2025).
- [1] I. Sepulveda and **B. Rachunok**. "Comparing online and offline Social Capital". In: *International Journal of Disaster Risk Reduction* (2025).

## Selected Honors & Awards

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### KIETS Climate Leaders Program

*Kenan Institute for Engineering, Technology & Science*

2026

Selected with Ph.D. student Juan Chala as a 2026 Student/Mentor team.

### Best Paper Award, Foundations of Risk

*Society for Risk Analysis*

2025

Award given to Ph.D. Student Yamil Essus's paper on ecological resilience.

### C.A. Anderson Outstanding Faculty Award

*NC State ISE Department*

2024

Award given to one faculty per year as voted by graduating seniors

### Sustainable Society Conference Delegate

*University Global Partnership Network*

2024

One of 8 faculty selected from NC State to attend

### Rising Environmental Leadership Program Fellow

*Stanford Woods Institute for the Environment*

2021

### Outstanding Research Award

*Purdue University School of Industrial Engineering*

2020

### Outstanding Service Award

*Purdue University School of Industrial Engineering*

2020

### Student Spotlight

*Society for Risk Analysis Engineering and Infrastructure Specialty Group*

2020

### Lee Chaden Fellowship

*Purdue University School of Industrial Engineering*

2018

### Estus and Vashti Magoon Award

*Purdue University*

2018

## Research Advising

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### Current PhD Student Advisees.....

**Ignacio Sepulveda**

*NC State University*

**Spring 2023-present**

*Industrial & Systems Engineering*

**Shakhawath Hossain**

*NC State University*

**Fall 2025-present**

*Industrial & Systems Engineering*

**Camryn Deames**

*NC State University*

Co-advised with Dr. Brandon McConnell

**Fall 2025-present**

*Business Management*

**Elizabeth Harris**

*NC State University*

**Fall 2025-present**

*Operations Research*

**Caroline Hayes**

*NC State University*

**Fall 2025-present**

*Operations Research*

**Juan Chala**

*NC State University*

**Fall 2025-present**

*Industrial & Systems Engineering*

### Current Master's Student Advisees.....

**Zackaria Radwan**

*NC State University*

**Fall 2026-present**

*Operations Research*

### PhD Alumni.....

**Yamil Essus**

*NC State University*

Current Position: Postdoctoral Fellow, University of Toronto

**Spring 2023-Spring 2026**

*Industrial & Systems Engineering*

**Robert Smith**

*NC State University*

Co-chaired with Dr. Brandon McConnell & Dr. Russell King. Current Position: Postdoctoral Research Associate, Institute for Risk Management and Insurance Innovation, UNC-Chapel Hill

**Spring 2024-Spring 2026**

*Operations Research*

**Austin Semmel**

*NC State University*

Co-chaired with Dr. Brandon McConnell & Dr. Sebastian Heese. Current Position: U.S. Military Academy at West Point

**Spring 2024-Spring 2026**

*Operations Research*

## Master's Alumni.....

**Teresa Groton**

*NC State University*

Current Position: U.S. Military Academy at West Point

**Fall 2024-Spring 2026**

*Operations Research*

**Patrick Leavitt**

*NC State University*

Current Position: Data Scientist, U.S. Coast Guard

**Fall 2023-Spring 2025**

*Industrial & Systems Engineering*

## NC State Graduate Advisory Committees.....

**Edric Pham**

*NC State University, Committee Member*

**Fall 2026-present**

*Graduate Advisory Committee*

**Madison Horgan**

*NC State University, Committee Member*

**Fall 2026-present**

*Graduate Advisory Committee*

**Ryan Andersen**

*NC State University, Committee Member*

**Summer 2026-present**

*Graduate Advisory Committee*

**Verónica Díaz-Pacheco**

*NC State University, Ph.D. Committee Member*

**Summer 2026-present**

*Operations Research*

**Cameron Lisy**

*NC State University, Ph.D. Committee Member*

**Summer 2026-present**

*Operations Research*

**Laura Mora**

*NC State University, Committee Member*

**Spring 2026-present**

*Graduate Advisory Committee*

**Roya Sahraei**

*NC State University, Committee Member*

**Spring 2026-present**

*Graduate Advisory Committee*

**Lindsey Jacobson**

*NC State University, Committee Member*

**Fall 2025-present**

*Graduate Advisory Committee*

**Yongseok Jeon**

*NC State University, Committee Member*

**Spring 2025-present**

*Graduate Advisory Committee*

**Lucia Rebeca Gomez Hurtado**

*NC State University, Committee Member*

**Spring 2025-present**

*Graduate Advisory Committee*

**Archit Sunil Kulkarni**

*NC State University, Committee Member*

**Spring 2025-present**

*Graduate Advisory Committee*

**Troy Wibowo**

*NC State University, Ph.D. Committee Member*

**Spring 2024-present**

*Industrial & Systems Engineering*

## Graduate School Representative Service.....

**Clayson Shumway**

*NC State University, Graduate School Representative*

**Summer 2025-present**  
*Economics & Management*

## External Graduate Advisory Committees.....

**Cameron Moore**

*Duke University, Ph.D. Committee Member*  
Advisor: Dr. Marta Zaniolo

**Spring 2026-present**  
*Graduate Advisory Committee*

**Swagato Biswas Ankon**

*University of Georgia, MS Committee Member*

**Spring 2023-present**  
*Environmental and Water Resources Engineering*

## Completed Graduate Advisory Committees.....

**Pranav Jain**

*NC State University, Ph.D. Committee Member*

**Spring 2023-Spring 2024**  
*Industrial & Systems Engineering*

## Graduate Independent Study Advisees.....

**Aakash Dhruv**

*NC State University*  
Current Position: Supply Chain Engineering, Amazon

**Spring 2024**  
*Integrated Manufacturing Systems Engineering Institute*

## Undergraduate Research Advisees.....

**Audrey Westlund**

*NC State University*

**Spring 2025-present**  
*Industrial & Systems Engineering*

**Ashton Sumpter**

*NC State University*

**Spring 2025-present**  
*Industrial & Systems Engineering*

**Matthew Bogdan**

*NC State University*

**Fall 2024-present**  
*Industrial & Systems Engineering*

**James Wilt**

*NC State University*

**Summer 2024-present**  
*Industrial & Systems Engineering REU*

**Ashelynn Jacobson**

*NC State University*

**Summer 2024-present**  
*Industrial & Systems Engineering REU*

**Drew Karnbad**

*NC State University*

**Spring 2024-present**  
*Industrial & Systems Engineering*

**Mike Mueller**

*NC State University*

**Spring 2024**  
*Industrial & Systems Engineering*

**Victoria Mendez Gonzalez**

*University of Puerto Rico at Mayagüez*

**Summer 2023**  
*Industrial & Systems Engineering*

**Fatemeh Heydari**  
NC State University

**Summer 2023**  
MI-REU Program

**Estevan Zamora**  
NC State University

**Summer 2023**  
MI-REU Program

**Julia Seidel**  
NC State University

**Spring 2023**  
Honors Thesis

**Adam Nayak**  
Stanford University

**2021-2022**

## Policy Engagement & Government Briefings

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**Fayetteville City Council Work Session**  
*North Carolina Data Center Water Resource Analysis*  
Invited policy briefing on data-center water, energy, wastewater, and infrastructure impacts

**Fayetteville, NC**  
August 2026

**North Carolina Energy Policy Task Force**  
*Initial Assessment Plan: North Carolina Data Center Water Resource Analysis*

**Raleigh, NC**  
July 2026

## Teaching

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ISE408: Design and Control of Production and Service Systems

Spring 2023-Fall 2025

ISE589: Computational Applications of Sustainability and Resilience

Fall 2023, Fall 2024, Fall 2025

## Invited Seminars

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**ECo Talks, The Earth Commons**  
*The Myth of Access: Affordability, Perception, and the Reality of Infrastructure Equity*

**Georgetown University**  
April 2026

**CEE Graduate Colloquium**  
*The Myth of Access: Quantifying Access to Essential Services Using AI, Durham, NC*

**Duke University**  
February 2026

**Civil and Environmental Engineering**  
*Risk, Optimization, and AI: Quantifying Access to Essential Services*

**Stanford University**  
February 2026

**Environmental, Water, and Coastal Engineering Seminar Series**  
*Quantifying Non-Spatial Dimensions of Access to Essential Services, Raleigh, NC*

**NC State University**  
December 2025

**Society for Risk Analysis Risk Science Webinar Series**  
*Quantifying and Operationalizing Resilience, Equality and Equity in Power Systems*

**Virtual**  
April 2025

**AI for Extreme Weather Event and Disaster Risk Management Seminar Series**  
*Quantifying Community Resilience*

**Virtual**  
October 2024

**Center for Environmental and Resource Economic Policy**  
*Engineering vs. Ecological Resilience*

**NC State University**  
April 2024



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| <b>International Risk and Resilience Research Alliance</b><br><i>The Pitfalls of Aggregation in Assessing Community Resilience</i> | <b>Virtual</b><br>February 2024            |
| <b>Energy Data Analytics Symposium</b><br><i>Ethics &amp; Social Responsibility of Data Analytics</i>                              | <b>Duke University</b><br>April 2023       |
| <b>Operations Research Seminar Series</b><br><i>Linking Sustainability &amp; Resilience with Operations Research</i>               | <b>NC State University</b><br>March 2023   |
| <b>Civil &amp; Environmental Engineering</b><br><i>Avoiding Resilience Traps with Unsupervised Learning</i>                        | <b>Cornell University</b><br>March 2022    |
| <b>Industrial &amp; Systems Engineering</b><br><i>Methods for Operationalizing Community Resilience Using Big Data</i>             | <b>NC State University</b><br>January 2022 |
| <b>INFORMS Student Chapter Summer Research Series</b><br><i>Social Resilience Fingerprinting</i>                                   | <b>Oklahoma University</b><br>April 2020   |

## Invited Conference Presentations

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| <b>Society for Risk Analysis Europe Annual Conference</b><br><i>Beyond Geography: Population-Scale Measurement of A-Spatial Food Access Using Large Language Models</i> | <b>Alicante, Spain</b><br>May 2026      |
| <b>Society for Risk Analysis Annual Meeting</b><br><i>Incorporating Equity and Fairness into a Mathematical Model for Power Restoration After Natural Hazards</i>       | <b>Washington, DC</b><br>December 2025  |
| <b>Society for Risk Analysis Annual Meeting</b><br><i>The Future of Risk Analysis: Perspectives from Emerging Risk Professionals</i>                                    | <b>Washington, DC</b><br>December 2025  |
| <b>INFORMS Annual Meeting</b><br><i>Incorporating Equity and Fairness into a Mathematical Model for Power Restoration After Natural Hazards</i>                         | <b>Atlanta, GA</b><br>October 2025      |
| <b>Society for Risk Analysis Europe Annual Conference</b><br><i>Post-disaster Power System Restoration Optimization</i>                                                 | <b>Stavanger, Norway</b><br>June 2025   |
| <b>Society for Risk Analysis Annual Meeting</b><br><i>Integrating Theories of Community Resilience</i>                                                                  | <b>Austin, TX</b><br>December 2024      |
| <b>INFORMS Annual Meeting</b><br><i>Predicting and Understanding Residential Water use with Interpretable Machine Learning</i>                                          | <b>Phoenix, AZ</b><br>October 2023      |
| <b>The International Symposium on Sustainable Systems and Technology</b><br><i>Quantifying Community Resilience to Break Resilience Traps</i>                           | <b>Fort Collins, CO</b><br>June 2023    |
| <b>Institute for Industrial and Systems Engineers Annual Conference</b><br><i>Quantifying Community Transformation</i>                                                  | <b>New Orleans, LA</b><br>May 2023      |
| <b>AGU Fall Meeting</b><br><i>Socio-hydrological impacts of water scarcity on household water affordability</i>                                                         | <b>New Orleans, LA</b><br>December 2022 |

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| <b>Society for Risk Analysis Annual Meeting</b><br><i>Socio-hydrological impacts of water scarcity on household water affordability</i>                                                   | <b>Tampa, FL</b><br>December 2022        |
| <b>Society for Risk Analysis Annual Meeting</b><br><i>Quantifying Community Resilience to Break Resilience Traps</i>                                                                      | <b>Virtual</b><br>December 2020          |
| <b>International Conference on Computational Social Science (IC2S2).</b><br><i>Be Careful Where You Get Your Tweets: Bias in Analysis from Tweet Acquisition Source</i>                   | <b>Virtual</b><br>June 2020              |
| <b>INFORMS Annual Meeting</b><br><i>Quantifying Community Transformation</i>                                                                                                              | <b>Virtual</b><br>October 2020           |
| <b>Natural Hazards Center Researcher's Meeting</b><br><i>Natural Hazard Twitter Analysis Depends on Tweet Sources</i>                                                                     | <b>Virtual</b><br>May 2020               |
| <b>INFORMS Annual Meeting</b><br><i>Social Resilience Fingerprinting</i>                                                                                                                  | <b>Seattle, WA</b><br>October 2019       |
| <b>Institute for Industrial and Systems Engineers Annual Conference</b><br><i>Interdependent Infrastructure System Risk &amp; Resilience to Natural Hazards.</i>                          | <b>Orlando, FL</b><br>May 2019           |
| <b>Society for Risk Analysis Annual Meeting</b><br><i>Electric Power Infrastructure Resilience</i>                                                                                        | <b>New Orleans, LA</b><br>December 2018  |
| <b>INFORMS Annual Meeting</b><br><i>The Sensitivity of Electric Power Infrastructure Resilience to the Spatial Distribution of Disaster Impacts</i>                                       | <b>Phoenix, AZ</b><br>October 2018       |
| <b>IEEE International Conference on Probabilistic Methods Applied to Power 2018 Systems</b><br><i>Stochastic Unit Commitment Performance Considering Monte Carlo Wind Power Scenarios</i> | <b>Boise, ID</b><br>June 2018            |
| <b>Society for Risk Analysis Annual Meeting</b><br><i>Transportation Network Recovery Analysis, an Equilibrated Perspective</i>                                                           | <b>Crystal City, VA</b><br>December 2017 |
| <b>INFORMS Annual Meeting</b><br><i>The Cost of Wrong Assumptions in Tropical Cyclone Frequency Distributions</i>                                                                         | <b>Houston, TX</b><br>October 2017       |
| <b>Institute for Industrial and Systems Engineers Annual Conference</b><br><i>UAVs Provide Lifesaving Medical Care</i>                                                                    | <b>Anaheim, CA</b><br>May 2016           |

## Student Conference Presentations

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Underline indicates an advisee

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| <b>Society for Risk Analysis Annual Meeting</b><br><i>Characterization of the Dependency Between Power and Communication Outages</i><br><u>E. Harris</u> and <b>B. Rachunok</b>                             | <b>Washington, DC</b><br>December 2025 |
| <b>Society for Risk Analysis Annual Meeting</b><br><i>Expanding the Dimensions of Food Access: Linking Experience and Geography Through Data-Driven Analysis</i><br><u>T. Groton</u> and <b>B. Rachunok</b> | <b>Washington, DC</b><br>December 2025 |

**93rd Military Operations Research Society Symposium**  
*A Framework for Analyzing Operational Efficiency in US Army Aviation*  
*Self-Organizing Map-based Clustering of Flight Dispatch Decisions*  
A. Semmel, H. S. Heese, B. McConnell, M. Powell, and **B. Rachunok**  
Selected Best in Working Group (Readiness)

**Leesburg, VA**  
June 2025

**Society for Risk Analysis Europe Annual Conference**  
*Quantifying Ecological Resilience*  
Y. Essus and **B. Rachunok**

**Stavanger, Norway**  
June 2025

**Society for Risk Analysis Annual Meeting**  
*Fairness in Post-Disaster Power Restoration*  
I. Sepulveda and **B. Rachunok**

**Austin, TX**  
December 2024

**Society for Risk Analysis Annual Meeting**  
*Quantifying Ecological Resilience*  
Y. Essus and **B. Rachunok**

**Austin, TX**  
December 2024

**INFORMS Annual Meeting**  
*Fairness in Post-Disaster Power Restoration*  
I. Sepulveda and **B. Rachunok**

**Seattle, WA**  
October 2024

**INFORMS Annual Meeting**  
*Estimating Access risk during blackouts due to vehicle electrification in North Carolina*  
Y. Essus and **B. Rachunok**

**Seattle, WA**  
October 2024

**International Symposium on Sustainable Systems and Technology**  
*Quantifying Social Capital*  
I. Sepulveda and **B. Rachunok**

**Baltimore, MD**  
June 2024

**International Symposium on Sustainable Systems and Technology**  
*Electric Vehicles Limit Equitable Access to Essential Services During Blackouts*  
Y. Essus and **B. Rachunok**

**Baltimore, MD**  
June 2024

**Society for Risk Analysis Annual Meeting**  
*Quantifying Social Capital*  
I. Sepulveda and **B. Rachunok**

**Washington, DC**  
December 2023

**Society for Risk Analysis Annual Meeting**  
*Electric Vehicles Limit Equitable Access to Essential Services During Blackouts*  
Y. Essus and **B. Rachunok**

**Washington, DC**  
December 2023

**INFORMS Annual Meeting**  
*Electric Vehicles Limit Equitable Access to Essential Services During Blackouts*  
Y. Essus and **B. Rachunok**

**Phoenix, AZ**  
October 2023

## Scientific Community Service

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**Reviewer**.....

### Journal Articles

*Communications Engineering, Journal of Infrastructure Systems, Journal of Water Resources Planning and Management, Nature Communications, IEEE Access, Risk Analysis, International Journal of Disaster Risk Reduction, Reliability Engineering*

*and System Safety, Utility Policy, Journal of Management in Engineering*

#### External Grant Proposals

DOE Science Synthesis Prize, November 2024

NSF Human-Environment Geographical Sciences Program, ad hoc, May 2023

NSF Human-Environment Geographical Sciences Program, ad hoc, January 2023

#### **Conference Organizing** .....

##### **Sustainability and Resilience of Infrastructure Track Chair**

*International Symposium on Sustainable Systems and Technology (ISSST)* 2024

##### **Equity, Sustainability and Engineering Session Chair**

*Society for Risk Analysis Annual Meeting* 2023

#### **Professional Societies** .....

*Society for Risk Analysis* 2016-present

Council member 2021-2023, Secretary 2024-present

INFORMS 2017-present

*Institute for Industrial and Systems Engineers* 2016-present

#### **University Service** .....

#### **Department and University Committees** .....

*ISE Masters of Engineering Management Committee* Fall 2023, Spring 2024

*ISE PhD Admissions Committee* Spring 2023, Fall 2024, Spring 2025

#### **Other Service** .....

*Engineering Open House* Spring 2023, Fall 2023, Spring 2024, Fall 2024

*ISE Summer Camp* Summer 2023, Summer 2024